

OPERATING SYSTEMS

Process Management and Scheduling

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- Not an independent scheduling algorithm
- Make use of other existing algorithms to group
- Multiple queues are maintained for processes with common characteristics
- Each queue can have its own scheduling algorithms
- Priorities are assigned to each queue
- CPU-bound jobs can be scheduled in one queue and all I/O-bound jobs in another queue
- Process Scheduler then alternately selects jobs from each queue and assigns them to the CPU based on the algorithm assigned to the queue

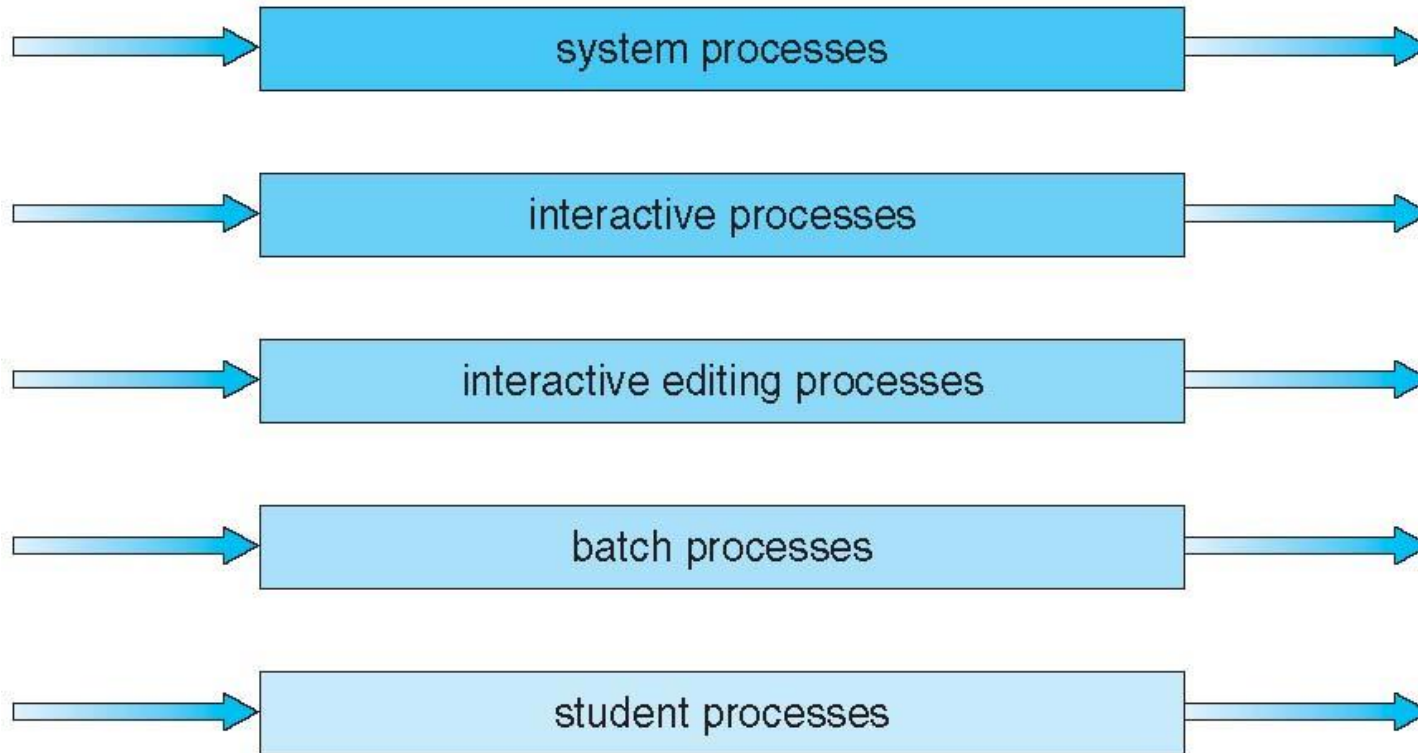
- Processes are **classified into groups**, for example:
 - **foreground** (interactive) processes
 - **background** (batch) processes
- Ready queue is partitioned into separate queues, e.g.:
 - **Foreground** and **background** queues
- Process is permanently assigned to one queue
- Each queue has its own scheduling algorithm, e.g.:
 - foreground – RR
 - background – FCFS

- Scheduling must be done between the queues:
 - **Fixed priority** scheduling
 - For example, foreground queue might have absolute priority over background queue
 - Serve all from foreground then from background
 - May lead to starvation
 - **Time slice** scheduling
 - Each queue gets a certain amount of CPU time which it can schedule amongst its processes, e.g.:
 - 80% to foreground in RR
 - 20% to background in FCFS
- **Advantage:** Low scheduling overhead
- **Disadvantage:** Inflexible

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Multi Level Queue Scheduling

highest priority



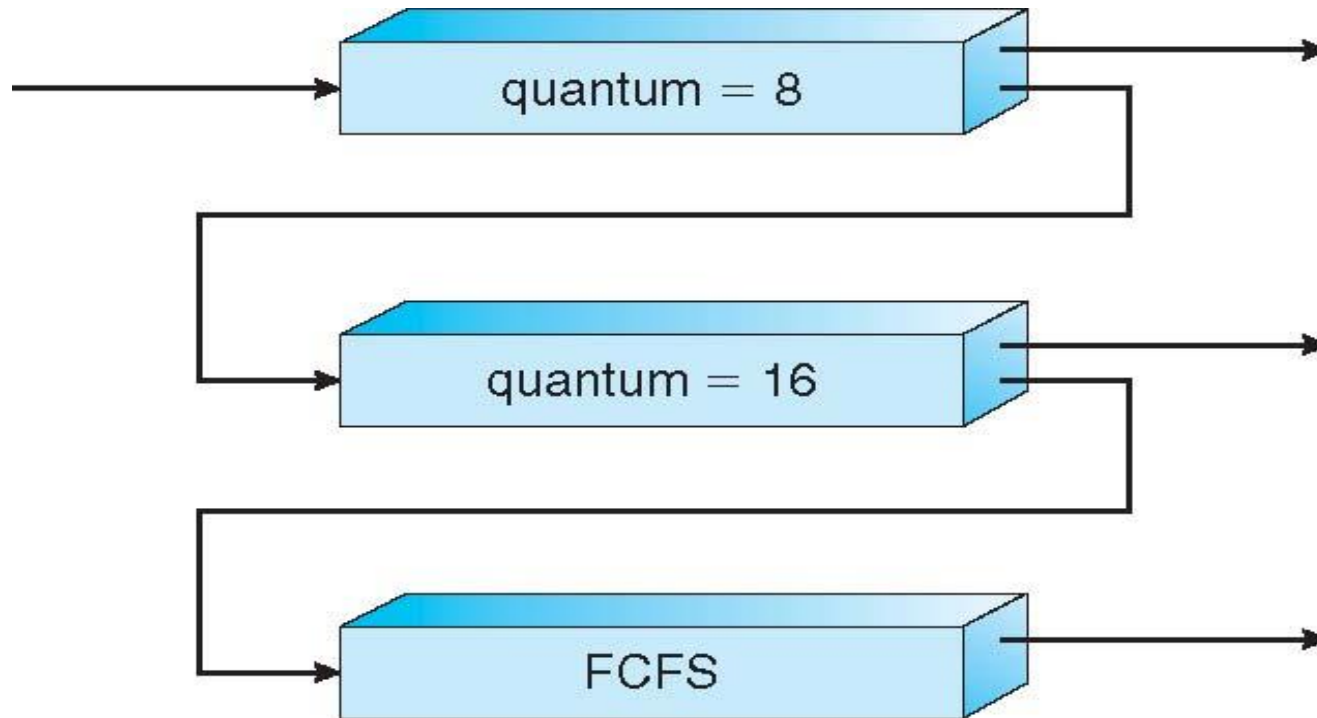
lowest priority

- A process can move between the various queues; aging can be implemented this way
- Multilevel-feedback-queue scheduler defined by the following parameters:
 - number of queues
 - scheduling algorithms for each queue
 - method used to determine when to upgrade a process
 - method used to determine when to demote a process
 - method used to determine which queue a process will enter when that process needs service

- Three queues:
 - Q_0 – RR with time quantum 8 milliseconds
 - Q_1 – RR time quantum 16 milliseconds
 - Q_2 – FCFS
- Scheduling
 - A new job enters queue Q_0 which is served FCFS
 - When it gains CPU, job receives 8 milliseconds
 - If it does not finish in 8 milliseconds, job is moved to queue Q_1
 - At Q_1 job is again served FCFS and receives 16 additional milliseconds
 - If it still does not complete, it is preempted and moved to queue Q_2

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Multi Level Feedback Queue Scheduling





THANK YOU

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